

Nasir U. Eisty

Assistant Professor
Department of EECS
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Research Interests

- Software Engineering for Science
- AI for Software Engineering
- Empirical Software Engineering
- Research Software Engineering
- Software Quality
- Software Security

Experience

- **Assistant Professor - Tenure Track** (August 2025 to Present)
Department of Electrical Engineering and Computer Science
The University of Tennessee, Knoxville, TN, USA
- **Assistant Professor - Tenure Track** (August 2021 to July 2025)
Department of Computer Science
Boise State University, Boise, ID, USA
- **Assistant Professor - Tenure Track** (September 2020 to July 2021)
Dept. of Computer Science & Software Engineering
California Polytechnic State University, San Luis Obispo, CA, USA
- **Postdoctoral Researcher** (June 2020 to August 2020)
Software Engineering lab, Dept. of Computer Science
University of Alabama, Tuscaloosa, AL, USA
- **Research Assistant** (August 2016 to May 2020)
Software Engineering lab, Dept. of Computer Science
University of Alabama, Tuscaloosa, AL, USA
- **Research Intern** (Summer 2019)
Los Alamos National Laboratory (LANL)
Los Alamos, NM, USA
- **Research Intern** (Summer 2017)
National Center for Supercomputing Applications (NCSA)
University of Illinois at Urbana-Champaign, IL, USA
- **Lecturer** (January 2015 to July 2016)
Department of Computer Science & Engineering.
Daffodil International University, Dhaka, Bangladesh

Education

- **Ph.D. in Computer Science (May 2020)**
Department of Computer Science
The University of Alabama, Tuscaloosa, AL, USA
- **MS in Computer Science (August 2018)**
Department of Computer Science
The University of Alabama, Tuscaloosa, AL, USA
- **B.Sc. in Computer Science & Engineering (December 2014)**
Department of Computer Science & Engineering
Rajshahi University of Engineering & Technology
Rajshahi, Bangladesh

Awards & Grants

- **FMitF: Track III: Teaching Program Specifications to the Scientific Computing Community**
 - Awarded grant \$248,779 from NSF, Duration 2026-2028
- **Code Review in Research Software - The RSE Experience. Role: Co-PI**
 - Awarded grant \$250,000 from the Sloan Foundation, Duration: 2023 - 2025
- **Research, Economic Development and Graduate Education (R-EDGE) at Cal Poly**
 - Awarded grant \$18,000 for the calendar year 2021
- **BSSw (Better Scientific Software) Fellowship 2020**
 - Awarded grant \$35,489 for the calendar year 2020
- **US-RSE Community Travel Grant to Attend Dagstuhl Seminar in Germany - 2024**
- **US-RSE Travel Grant to Attend Research Software Community Building Workshop at Princeton University - 2022**
- **Travel Grant to Attend DOE ECP Annual Meeting at Houston, Texas for Multiple Years - 2019 to 2023**
- **1st Prize in Graduate Poster Competition at the University of Alabama - 2019**
- **BSSw (Better Scientific Software) Fellowship Honorable Mention - 2019**
- **Travel Scholarship to Einstein Toolkit Meeting 2017**

Publications

- Gabrielle O'Brien, Nasir U. Eisty “**Scientific Software in the Age of Vibe Coding**” Computing in Science & Engineering, vol. 28, no. 1, pp. 112–116, Jan.–Mar. 2026.
- Nasir Eisty, Todd Evans “**Bridging Two Worlds: Introducing the Software Engineering and Scientific Computing Department**” Computing in Science & Engineering, 2026.
- Nasir U. Eisty, Jeffrey C. Carver, Jamil Cohoon, Ian A. Cosden, Carole Goble, Sandra Grayson, et al. “**Ten Simple Rules for Catalyzing Collaborations and Building Bridges**”

between Research Software Engineers and Software Engineering Researchers”
Computing in Science & Engineering, 2025.

- Stephan Druskat, Nasir U. Eisty, Robert Chisholm, Neil Chue Hong, Richard C. Cocking, Maximilian B. Cohen, et al. **“Better Architecture, Better Software, Better Research”**
Computing in Science & Engineering, 2025.
- Jerney Husle, Nasir Eisty, Tim Menzies **"Shaky Structures: The Wobbly World of Causal Graphs in Software Analytics"** Empirical Software Engineering Journal (EMSE), June 2025
- Shane Panter, Lucas Hindman, Nasir Eisty **"PVAC: package version activity categorizer, leveraging semantic versioning in a heterogeneous system"** Empirical Software Engineering Journal (EMSE), May 2025
- Nasir Eisty, Upulee Kenewala, Jeffrey Carver **"Testing Research Software: An In-Depth Survey of Practices, Methods, and Tools"** Empirical Software Engineering Journal (EMSE), January 2025
- Sharif Ahmed, Nasir Eisty **"Hold On! Is My Feedback Useful? Evaluating the Usefulness of Code Review Comments"** Empirical Software Engineering Journal (EMSE), January 2025
- Sheikh Md. Mushfiqur Rahman, Gregory Watson, Nasir Eisty **“Exploring Sustainability in Scientific Software through Code Quality & Test Coverage Metrics”** *Platform for Advanced Scientific Computing (PASC) 2026, Bern, Switzerland.*
- Eric Melin, Nasir Eisty, Gregory Watson, Addi Malviya Thakur **“Multi-Artifact Analysis of Self-Admitted Technical Debt in Scientific Software”** *Platform for Advanced Scientific Computing (PASC) 2026, Bern, Switzerland.*
- Sharif Ahmed, Nasir Eisty **“Characterizing the Usefulness of Code Review Comments in Scientific Software for Software Quality and Scientific Rigor”** *Platform for Advanced Scientific Computing (PASC) 2026, Bern, Switzerland.*
- Eric Melin, Nasir Eisty, Addi Thakur, Greg Watson **“Self-Admitted Technical Debt in Scientific Software: Prioritization, Sentiment, and Propagation Across Artifacts”** *Practice and Experience in Advanced Research Computing (PEARC) 2026, Minneapolis, MN, USA.*
- Ariful Malik, Jeffrey Carver, Nasir Eisty **“Peer Code Review in Research Software: Practices, Improvements, and Implementation Challenges from Interviews of RSEs”** *Practice and Experience in Advanced Research Computing (PEARC) 2026, Minneapolis, MN, USA.*
- Shane Panter, Nasir Eisty **“Technical Lag as Latent Technical Debt: A Rapid Review”** *International Conference on Technical Debt (TechDebt) 2026.*
- Saznin Sultana, Sadia Afreen, Nasir Eisty **“LLMs in Code Vulnerability Analysis: A Proof of Concept”** *4th International Workshop on Software Vulnerability Management (SVM) 2026.*
- James A. Curtis, Nasir U. Eisty **“The Kubernetes Security Landscape: AI-Driven Insights from Developer Discussions”** *23rd IEEE/ACIS International Conference on Software Engineering Research, Management, and Applications (SERA 2025).*

- Eric L. Melin, Nasir U. Eisty **“Exploring the Advances in Using Machine Learning to Identify Technical Debt and Self-Admitted Technical Debt”** *23rd IEEE/ACIS International Conference on Software Engineering Research, Management, and Applications (SERA 2025)*.
- Sheikh Md. Mushfiqur Rahman, Nasir U. Eisty **“Introducing Ensemble Machine Learning Algorithms for Automatic Test Case Generation Using Learning-Based Testing”** *23rd IEEE/ACIS International Conference on Software Engineering Research, Management, and Applications (SERA 2025)*.
- Sharif Ahmed, Rey Ortiz, Nasir Eisty **“Decade-long Utilization Patterns of ICSE Technical Papers and Associated Artifacts”** *22nd IEEE/ACIS International Conference on Software Engineering Research, Management, and Applications (SERA 2024), May 30–June 1, 2024, Honolulu, USA*.
- Shariful Alam, Jidong Xiao, Nasir Eisty **“Exploiting CPU Clock Modulation for Covert Communication Channel”** *22nd IEEE/ACIS International Conference on Software Engineering Research, Management, and Applications (SERA 2024), May 30–June 1, 2024, Honolulu, USA*.
- Shane Panter, Nasir U. Eisty, **“Rusty Linux: Advances in Rust for Linux Kernel Development”** *18th ACM/IEEE International Symposium on Empirical Software Engineering and Measurement (ESEM 2024), October 24-25, Barcelona, Spain*
- Sharif Ahmed, Nasir U. Eisty, **“Understanding Emojis :) in Useful Code Review Comments”** *NL-based Software Engineering (NLBSE) co-located at 46th International Conference on Software Engineering (ICSE 2024), April 20, Lisbon, Portugal*
- Sharif Ahmed, Nasir U. Eisty, **“Exploring the Advances in Identifying Useful Code Review Comments”** *17th ACM/IEEE International Symposium on Empirical Software Engineering and Measurement (ESEM 2023), October 26-27, New Orleans, USA*
- Jason Duran, Mostofa Najmus Sakib, Nasir U. Eisty, Francesca Spezzano. **“Evaluating Code Metrics in GitHub Repositories Related to Fake News and Misinformation”** *21st IEEE/ACIS International Conference on Software Engineering Research, Management, and Applications (SERA 2023), May 23-25, Orlando, USA*
- Jeffrey Fairbanks, Akshharaa Tharigonda, Nasir U. Eisty, **“Analyzing the Effects of CI/CD on Open Source Repositories in GitHub and GitLab”** *21st IEEE/ACIS International Conference on Software Engineering Research, Management, and Applications (SERA 2023), May 23-25, Orlando, USA*
- Patrick Chadbourne, Nasir U. Eisty, **“Applications of Causality and Causal Inference in Software Engineering”** *21st IEEE/ACIS International Conference on Software Engineering Research, Management, and Applications (SERA 2023), May 23-25, Orlando, USA*
- Bishal Lakha, Kalyan Bhetwal, Nasir U. Eisty, **“Analysis of Software Engineering Practices in General Software and Machine Learning Startups”** *21st IEEE/ACIS International Conference on Software Engineering Research, Management, and Applications (SERA 2023), May 23-25, Orlando, USA*
- Md Athikul Islam, Rizbanul Hasan, Nasir U. Eisty, **“A Systematic Literature Review on Optimizing Documentation Effort in Agile Software Development”** *21st IEEE/ACIS*

International Conference on Software Engineering Research, Management, and Applications (SERA 2023), May 23-25, Orlando, USA

- Nasir U. Eisty, Jeffrey C. Carver **“Testing of Research Software: A Survey”** *Empirical Software Engineering Journal* 2022
- Sharif Ahmed, Arif Ahmed, Nasir U. Eisty, **“Automatic Transformation of Natural to Unified Modeling Language: A Systematic Review”** *20th IEEE/ACIS International Conference on Software Engineering Research, Management, and Applications (SERA 2022), May 25-27, Las Vegas, USA*
- Shashank B. Lakshman, Nasir U. Eisty, **“Software Engineering Approaches for TinyML based IOT Embedded Vision: A Systematic Literature Review”** *4th International Workshop on Software Engineering Research & Practices for the Internet of Things (SERP4IoT 2022) Colocated with ICSE 2022, May 10, 2022*
- Nasir U. Feisty, Jeffrey C. Carver, **“Developers Perceptions of Peer Code Review in Research Software Development”** *Empirical Software Engineering Journal* 27, 13 (2022)
- Nasir U. Eisty, Danny Perez, Jeffrey C. Carver, J. David Moulton, Hai Ah Nam **“Testing Research Software: A Case Study”** *International Workshop on Software Engineering for Computational Science at Amsterdam, Netherlands, June 3-5, 2020.*
- Nasir U. Eisty, George K. Thiruvathukal, Jeffrey C. Carver **“Use of Software Process in Research Software development: A Survey”** *In Evaluation and Assessment in Software Engineering (EASE'19) Conference at Copenhagen, Denmark, April 15 - 17, 2019.*
- Nasir U. Eisty, George K. Thiruvathukal, Jeffrey C. Carver **“A Survey of Software Metric Use in Research Software Development”** *14th eScience IEEE International Conference at Amsterdam, Netherlands, Oct 29 - Nov 1, 2018.*

Posters

- Alex Koufos, Nasir Eisty, Miranda Mundt **“Software Testing Practices for Reproducible Open Science”** at AGU 2024, Washington D.C., December 9 - 13, 2024
- Nasir Eisty, Myra B. Cohen, Neil Chue Hong, Ryan C. Cocking, Stephan Druskat, Michael Felderer, Samuel Grayson, Lars Grunske, Wilhelm Hasselbring, Sarah Harris, Jan Linxweiler, and Colin C. Venters, **“Better Architecture, Better Software, Better Research”**, US-RSE Conference, Albuquerque, NM, USA October 15 - 17, 2024

Students

- **MS - Shaznin Sultana** (Graduated in May 2025)
- **PhD - Shariful Alam** (Graduated in May 2025)
- **PhD - Sharif Ahmed** (Graduated in June 2025)
- **PhD - Shane Panter** (Graduated in May 2026)
- **PhD - Eric Melin** (Expected Graduation May 2026)
- **PhD - Alex Curtis** (Expected Graduation July 2027)

Invited Talks & Participation

- **Dagstuhl Seminar: Research Software Engineering: Bridging Knowledge Gaps ([Link](#))**
Held from April 14 to April 19, 2024, at Schloss Dagstuhl in Wadern, Germany.
- **Webinar: Testing and Code Review Practices in Research Software Development ([Link](#))**
The IDEAS Productivity Webinar Series of DOE Exascale Computing Project
September 9, 2020
- **Collaboration Opportunities with the 2020 BSSw Fellows**
2021 Exascale Computing Project Annual Meeting
April 13, 2021(Virtual)
- **Automatic Testing for Scientific Software Development ([Link](#))**
SIAM Conference on Computational Science and Engineering (CSE 21)
March 01, 2021 (Virtual)
- **Guest Lecture on Peer Code Review Practices**
Spring 2021, Ball State University, Muncie, IN, USA

Short Courses Taught

- **Tutorial: Contemporary Peer Code Review Practices**
2019 Exascale Computing Project Annual Meeting
January 17, 2019, at Houston, Texas
- **Tutorial: Contemporary Peer Code Review in Scientific Software Development ([Link](#))**
14th eScience IEEE International Conference
October 29, 2018, at Amsterdam, Netherlands
- **Summer Camp: Student Introduction to Engineering (Computer Science)**
SITE III Summer Camp
July 22 - 27, 2018 at The University of Alabama
- **Summer Camp: Student Introduction to Engineering (Computer Science)**
SITE II Summer Camp
July 15 - 20, 2018 at The University of Alabama
- **Summer Camp: Student Introduction to Engineering (Computer Science)**
SITE I Summer Camp
July 8 - 13, 2018 at The University of Alabama
- **Tutorial: Contemporary Peer Code Review, A Lightweight, Asynchronous Method for Ensuring High-Quality Code**
SEA Software Engineering Conference and Tutorial
April 4, 2018 at UCAR - Boulder, CO, USA

Workshops Organized

- **Scientific Software and the People who Make it Happen: Building Communities of Practice**
The International Conference for High Performance Computing, Networking, Storage, and Analysis (SC32) ([Link](#))
November, 2023
- **Software Engineering and Reuse in Modeling, Simulation, and Data Analytics for Science and Engineering**

The International Conference for High Performance Computing, Networking, Storage, and Analysis (SC22) ([Link](#))

November, 2022

- **Software Engineering and Reuse in Modeling, Simulation, and Data Analytics for Science and Engineering** ([Link](#))

The International Conference for High Performance Computing, Networking, Storage, and Analysis (SC21)

November 16, 2021 (Virtual)

- **Software Engineering and Reuse in Modeling, Simulation and Data Analytics for Science and Engineering** ([Link](#))

The International Conference for High Performance Computing, Networking, Storage, and Analysis (SC20)

November 19, 2020 (Virtual)

- **2019 International Workshop on Software Engineering for Science** ([Link](#))

International Conference on Software Engineering (ICSE)

May 28, 2019 at Montreal, Canada

- **Understanding Challenges and Solutions for the Development of Modeling & Simulation Software in Defense and Medicine**

AlaSim International Conference

October 25, 2017 at Huntsville, AL, USA

Professional Services

- **BoF Track Co-chair**

United States Research Software Engineering Conference 2024 (US-RSE 24)

October 15-17, 2024 at Albuquerque, New Mexico, USA

- **Program Committee Member**

International Symposium on Empirical Software Engineering and Measurement (ESEM 24)

October 24-25, 2023 at Barcelona, Spain

- **Program Committee Member**

United States Research Software Engineering Conference 2023 (US-RSE 23)

October 16-18, 2023 at Chicago, Illinois, USA

- **Program Committee Member**

International Symposium on Empirical Software Engineering and Measurement (ESEM 23)

October 23-27, 2023 at New Orleans, Louisiana, USA

- **Program Committee Member**

52nd International Conference on Parallel Processing (ICPP)

August 7-10, 2023 at Salt Lake City, Utah, USA

- **Program Committee Member**

Research Software Engineers in eScience (RSE-eScience-2023)

October 9, 2023 at Limassol, Cyprus, USA

- **Program Committee Member**

Software Engineering for Research Software (SE4RS'23)

July 23-27, 2023 at Portland, Oregon, USA

- **Journal Reviewer**

Empirical Software Engineering (EMSE)

January 2023 - Present

➤ **Journal Reviewer**

IEEE Transactions of Software Engineering (TSE)

January 2022 - Present

➤ **Program Committee Member ([Link](#))**

2022 International Workshop on Software Engineering for eScience

October 11, Salt Lake City, Utah

➤ **Program Committee Member ([Link](#))**

Research Software Engineers in eScience (RSE-eScience-2022)

October 11, Salt Lake City, Utah

➤ **Program Committee Member ([Link](#))**

US-RSE Virtual Workshop 2022

September 14th & 16th, 2022

➤ **Program Committee Member ([Link](#))**

2022 International Workshop on Software Engineering for Computational Science

June 21-23, 2022 at London, UK

➤ **Journal Reviewer**

Software Quality Journal

August 2021 - Present

➤ **Journal Reviewer**

Nature Scientific Reports

January 2022 - Present

➤ **PhD Admission Committee Member**

PhD in Computing, Boise State University

August 2021 - Present

➤ **Undergraduate Success Committee Member**

Dept. of Computer Science, Boise State University

August 2021 - Present

➤ **Program Committee Member ([Link](#))**

2021 International Workshop on Software Engineering for Computational Science

June 16-18, 2021 at Krakow, Poland

➤ **Software Engineering Lecturer Search Committee Member**

Fall 2020 in CSSE Department at Cal Poly

➤ **Scholarship Award Committee Member**

Fall 2020 - Winter 2021 in CSSE Department at Cal Poly

➤ **Software Engineering Curriculum Committee Member**

Fall 2020 - Winter 2021 in CSSE Department at Cal Poly

➤ **Research Software Engineer (RSE) Search Committee Member**

University of Alabama Research Software Initiative

Blogs

➤ **Recommendations for Peer Code Review in Research Software Development ([Link](#))**

Published on Better Scientific Software Blog by DOE on Sep 14, 2022

➤ **Informational Interview Questions for Forming an RSE Group ([Link](#))**

Published on US-RSE blog on June 04, 2022

➤ **Use of Software Metrics in Research Software ([Link](#))**

Published on Better Scientific Software Blog by DOE on April 10, 2021

➤ **Testing Non-Deterministic Research Software ([Link](#))**

Published on Better Scientific Software Blog by DOE on October 13, 2020

➤ **Making Software a First-Class Citizen in Research ([Link](#))**

Software Sustainability Institute Blog on November 28, 2018

Notable Participations

➤ **Mid-semester Course Assessment Training by CTL at Boise State**

September 11, 2024 at Boise State University

➤ **Course Design Institute (CDI) by CTL at Boise State**

May 16 - 20, 2022 at Boise State University

➤ **US-RSE Community Building Workshop**

April 25 - 26, 2022 at Princeton University

➤ **Judge at 2022 Graduate Student Showcase Awards**

At Boise State University

➤ **University of Alabama System Scholars Institute 2019**

May 14, 2019 at University of Alabama in Huntsville, AL, USA

➤ **Judge at Alabama Robotics Competition**

April 6, 2019 at University of Alabama, Tuscaloosa, AL, USA

➤ **Judge at Undergraduate Research and Creative Activity Conference 2019**

March 27, 2019 at University of Alabama, Tuscaloosa, AL, USA

➤ **ALEPSCOR NSF Day**

May 1, 2018 at University of Alabama at Birmingham, AL, USA

➤ **Judge at Alabama Robotics Competition**

April 7, 2018, at The University of Alabama

➤ **Judge at 2018 Undergraduate Research and Creative Activity Conference**

March 28-30, 2018, at The University of Alabama

➤ **2018 Alabama Computer Science Education Summit**

March 14, 2018, Montgomery, AL, USA

➤ **2017 International Workshop on Software Engineering for High Performance Computing in Computational and Data-Enabled Science and Engineering**

SC17 International Conference, November 12, 2017, Denver, CO, USA

➤ **2017 North American Einstein Toolkit School and Workshop**

Jul 31 - Aug 4 at National Center for Supercomputing Applications, Urbana, IL, USA

➤ **Scaling to Petascale Institute - BLUE WATERS Sustained Petascale Computing**

June 26 - 30, 2017 at Beckman Institute for Advanced Science and Technology, Urbana, IL

➤ **Volunteer at Alabama Robotics Competition**

April 8, 2017 at University of Alabama, Tuscaloosa, AL, USA

Others

- **Excellent Performance Award** from Dept. of CSE, RUET in 2013
- **Excellent Performance Award** from Dept. of CSE, RUET in 2014
- **Champion** prize at inter university software/project contest at BUET CSE-FEST 2013
- **Champion** prize in software category at digital innovation fair 2013.
- **7th position** at EATL-Prothomalo Apps contest 2014
- **7th position** in IT Business Idea challenge at IUT CSE Carnival 2014
- **5th Position** at Inter-University Programming Contest at RUET CSE-FEST in 2012
- **Runner up** at programming contest at RUET in 2011
- **Jury Board member** at “**National Mobile apps contest**” in Bangladesh for the year 2015